

OPERATIONAL EXPERIENCE OF SULPHUR RETENTION IN A BUBBLING FLUIDISED BED UNDER OXY-FUEL COMBUSTION CONDITIONS

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Abstract: Oxy-fuel combustion is one of the leading options for power generation with CO₂ capture. The process consists on burning the fuel with a mixture of nearly pure oxygen and a CO₂-rich recycled flue gas resulting in a product flue gas from the boiler containing mainly CO₂ and H₂O. Among the possible boiler types, fluidised bed (FB) combustors are very appropriate for the oxy-fuel process because allow the in-situ desulphurisation by feeding calcium-based sorbents into the combustor.

In this work, the effect of the combustion temperature on the retention of the SO₂ generated during the combustion of a Spanish anthracite has been analyzed. The experimental facility has been a bubbling fluidised bed (BFB) combustor of 3 kW_{th} operating under oxy-fuel combustion conditions. Several tests were also carried out under enriched-air combustion conditions for comparison reasons. A Spanish limestone “Granicarb” was used as Ca-based sorbent for sulphur retention. The temperatures tested were between 800 and 975 °C using different Ca/S molar ratios.

It was found that under oxy-fuel combustion conditions in BFB the optimum temperature to achieve the highest sulphur retention was 900-925 °C, whereas in enriched-air combustion the optimum was ≈850-870 °C. Working at the optimum temperature for each case, the SO₂ retentions were slightly lower in oxy-fuel combustion conditions than in the conventional air-fired combustion.

Keywords: Oxy-fuel combustion, SO₂ retention, limestone, fluidised bed.

INTRODUCTION

The emission of gases into the atmosphere from burning fossil fuels, mainly coal, in large power plants causes important environmental problems. On the one hand, the CO₂ from combustion has been recognized as one of the major contributors to the build-up of greenhouse gases in the atmosphere and, by the other hand, sulphur in coal is oxidised to SO₂ during combustion which contributes to acid rain formation.

The capture and storage of CO₂, emitted in large quantities from power plants, is considered an option to be explored in the medium term for reducing CO₂ levels released to the atmosphere. Oxy-fuel combustion is one of the possibilities under investigation within the different options for CO₂ capture (Wall et al., 2009; Kanniche et al., 2010; Toftegaard et al., 2010). This technology uses for combustion pure O₂ (instead of air used in conventional combustion) mixed with CO₂-rich recycled flue gas, and so, the flue gas stream from the boiler finally produced is highly concentrated on CO₂. After condensing water the CO₂ could be ready for purification and stored.

Among the possible boiler types, fluidised bed (FB) combustors are very appropriate for the oxy-fuel process because they allow the in-situ desulphurisation of the flue gas by feeding calcium-based sorbents into the combustor, such as limestone or dolomite. In-situ

desulphurisation can be a great advantage since sulphur containing species mean a risk of corrosion and can have impacts in the furnace, during ash collection, CO₂ compression, transport and storage (Stanger and Wall, 2011).

It is also believed that oxy-fuel circulating fluidised bed (CFB) combustors will be an important candidate for new coal fired power plants (Myöhänen et al., 2009; Czakiert et al., 2010; Jia et al., 2010) because the circulation of solids in the combustor can help to an effective control of the temperature. Other important advantage is the efficiency improvement mainly due to three features: the strong mixing in the furnace and longer residence times of solids that allows a good carbon burnout; the higher oxygen concentration in the combustor that allows to achieve a reduction in area of the CFB boiler by as much as 50% (Liljedahl et al., 2006; Stamatelopoulos et al., 2008; Normann et al., 2008) and the possibility of reduction the flue gas recycling and thus reducing the size of some subsequent equipments and the power consumption. Other well known advantages are the possibility of burning a wide range of fuels and the relatively low NO_x emissions achieved (Normann et al., 2008; Jia et al., 2010).

Operating conditions in the FB combustors can lead to important differences in the sorbent behaviour. Fig. 1 shows the thermodynamic equilibrium curve of CaCO₃ calcination. The operating conditions existing in FB for conventional combustion with air are characterised by low CO₂ concentrations (≈15%) and temperatures about 850 °C. These conditions lead to a previous sorbent calcination (R1) and to the sulphation of calcines (R2), i.e. indirect sulphation:



In oxy-fuel combustion, CO₂ concentration in the flue gas may be enriched up to values as high as 90%. Therefore, the sorbent can be surrounded by CO₂ concentrations ranging from 40 to 90 % during their stage in the FB reactor. Under so high CO₂ concentrations, the sorbent can behave in two ways depending on the temperature (see Fig. 1) and covers both calcining and non-calcining conditions. At 850 °C, the sulphur retention will be produced under direct sulphation (R3), being necessary higher temperatures to operate under indirect sulphation (R1 and R2).

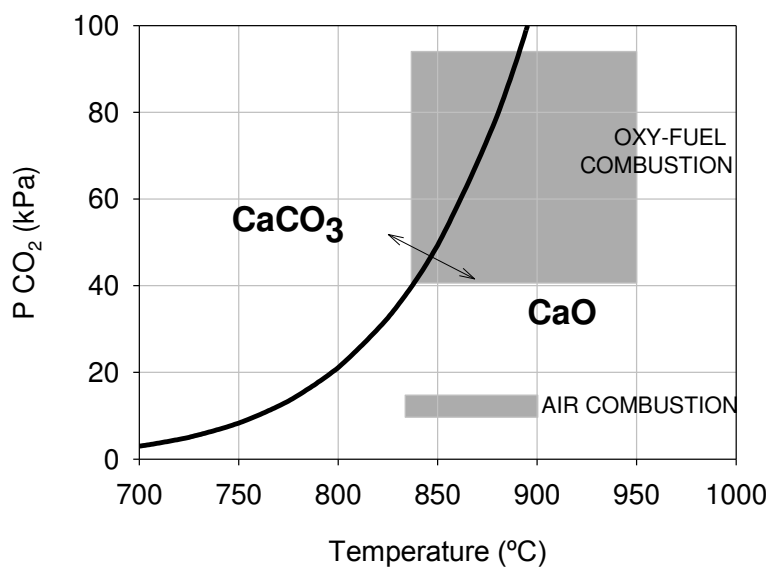


Fig. 1. Thermodynamic equilibrium curve of CaCO₃ calcination.

The objective of this work was to determine the optimum operating temperature in fluidised beds with respect to the sulphur retention working under oxy-fuel operating conditions. The experiments were carried out in a continuous bubbling fluidised bed (BFB) combustor (3 kW_{th}) operating under calcining and non-calcining conditions. A Spanish anthracite and a Spanish limestone were used in the tests. The effect of the main operating conditions with respect to the sorbent sulphation process was also analysed.

EXPERIMENTAL SECTION

Materials

The coal used for the combustion was a Spanish anthracite from León with a particle size of 0.2-1.2 mm. The calcium based sorbent used was a Spanish limestone, Granicarb, with a particle size of 0.3-0.5 mm, and characterised by its high purity. In addition, inert sand (0.2-0.6 mm) was fed together with the coal and the limestone during the tests to decrease the residence time of the sorbent in the fluidised bed. Table 1 shows the composition and main properties of the materials used.

Table 1. Composition and properties of coal and limestone.

Coal Anthracite		Sorbent Granicarb	
Composition (wt%)		Composition (wt%)	
Moisture	1.0	CaCO ₃	97.1
Ash	31.6	MgCO ₃	0.2
Volatiles	7.6	Na ₂ O	1.1
Fixed C	59.9	SiO ₂	<0.1
C	60.7	Al ₂ O ₃	<0.1
H	2.1	Fe ₂ O ₃	<0.1
N	0.9		
S	1.5	Porosity (%)	
		Raw	3.7
HHV (kJ/kg)	22363	Calcined	49

Experimental installation

Coal combustion experiments with limestone addition were carried out in a BFB combustor working in oxy-fuel combustion conditions. Tests under enriched-air conditions were also carried out for comparison reasons.

The experimental installation consists of a fluidised bed reactor, and different auxiliary systems for gas supply, solid feeding, and gas analysis. Fig. 2 shows a schematic diagram and photography of the installation.

The combustor consists of a stainless steel reactor of 9 cm i.d., 60 cm height and 50 cm of freeboard. The height of the bubbling fluidised bed was maintained constant at 40 cm. A heat exchanger located inside the bed allows us the control of temperature. The heat exchanger can be moved vertically through the reactor, and thus the heat extracted from the combustor is varied by modifying the contact surface inside the bed.

The gas supply system was based on mass flow controllers for air, CO₂ and O₂, to simulate typical gas compositions entering the reactor in oxy-fuel combustion conditions. A gas velocity of ≈40 cm/s (calculated at 850 °C) at the inlet of the bed was used during the tests. An air pre-heater allows introducing hot gases inside the reactor during the start-up of the plant.

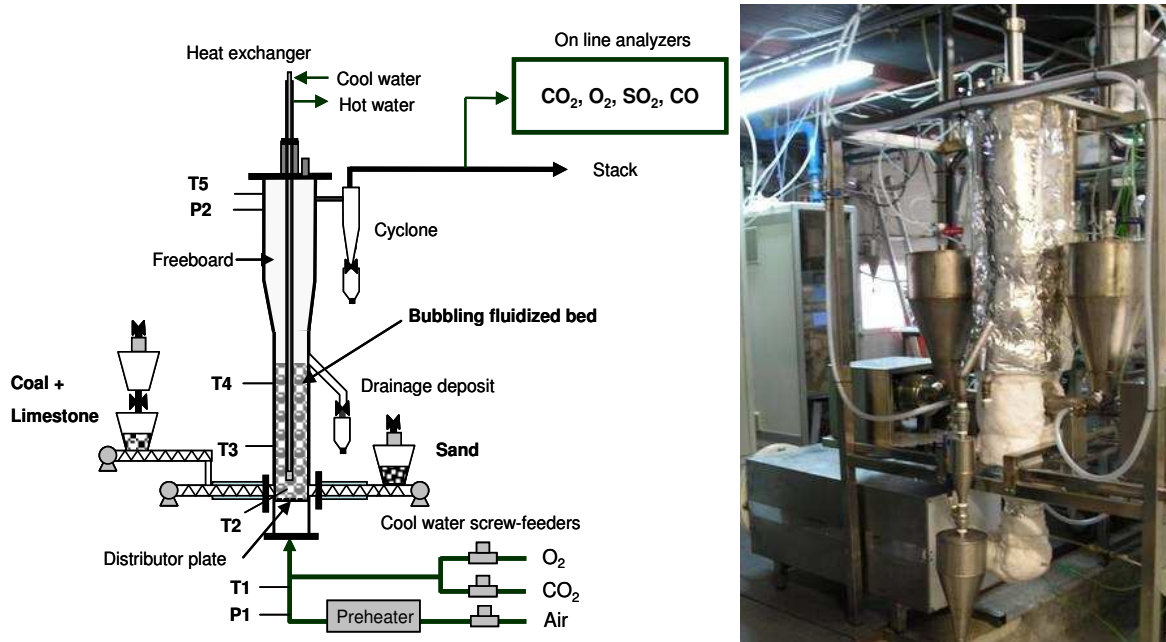


Fig. 2. Oxy-fuel Bubbling Fluidised Bed Combustor (3 kWth) at ICB-CSIC.

The solids were fed to the reactor by means of water-cooled screw feeders located just above the distributor plate. To assure a constant Ca/S molar ratio, coal and sorbent were fed together by means of two screw feeders in series: the first controls the solids feeding rate and the second one introduces the coal/sorbent mixture as quick as possible into the bed to avoid the coal pyrolysis and the plugging of the pipe.

An important feature of the tests carried out is the certainty that they are under steady state. Considering the fluidised bed dimensions and the limestone feeding rate, solid residence times of 10-20 hours would be normal for typical tests. To be sure that the steady state has been reached, very long operating times would be needed. To reduce the residence time of the sorbent inside the fluidised bed up to values of about 2 hours, inert silica sand was also fed into the reactor.

The flue gas stream passed through a high efficiency cyclone to recover the elutriated solids and then was sent to the stack. The composition of gases leaving the reactor was analysed continuously after water condensation by on-line gas analysers. CO₂, CO, and SO₂ were determined in a Non-Dispersive Infra-Red analyser (NDIR, Siemens/Ulramat 23), and the O₂ in a paramagnetic analyser (Siemens/Oxymat 6).

The facility was equipped with pressure sensors and thermocouples to know the pressure drop and temperatures, respectively, along the installation. General process data (temperatures, pressures, mass flow-meters, gas composition, etc.) were recorded in a computer continuously.

Sulphur retention (SR) was calculated taking into account the feed rate of coal ($F_{0,coal}$), its ultimate analysis ($X_{S,coal}$), the gas flow rate (Q), and the SO₂ concentration (C_{SO_2}) measured in the flue gas at the exit of the reactor, by the following equation:

$$SR(\%) = \frac{(F_{0,coal} X_{S,coal} / M_{SO_2}) - Q \cdot C_{SO_2}}{F_{0,coal} X_{S,coal} / M_{SO_2}} \cdot 100$$

The SO₂ concentration was considered as an average value of the obtained along the whole duration of the test under steady-state conditions.

RESULTS

The effect of the main variables affecting to the process of sorbent sulphation under oxy-fuel conditions was analysed in the BFB combustor. In addition, some experiments were also carried out under enriched-air conditions to establish a comparison between oxy-fuel and conventional combustion but using the same O_2 concentration.

The experiments were carried out at different temperatures between 800 and 975 °C and with an inlet O_2 concentration of 35 vol.%. CO_2 or N_2 were used to balance when working under oxy-fuel or enriched-air conditions, respectively. Two different Ca/S molar ratios (2-3) were used to determine the SR obtained in the reactor. Each test corresponds to a minimum of one hour under steady state conditions.

Influence of temperature.

To analyse the effect of temperature on the SR, several tests were carried out maintaining constant the Ca/S molar ratio and the O_2 concentration at the inlet. Fig. 3 shows the effect of temperature on SR working in oxy-fuel combustion conditions.

For the anthracite, SR initially increased with increasing temperature up to a maximum of 900-925 °C and then, further increases in temperature caused a decrease in SR values. These results are in good agreement with the results obtained in previous works where the sulphur retention capacity of the limestone “Granicarb” was analysed in a TGA and in a batch FB reactor under different conditions (García-Labiano et al., 2011; de Diego et al., 2011). In these works, it was found that the effect of the temperature was different depending on the sulphation pattern of the sorbent (see reactions R1-R3). In direct sulphation (non-calcining conditions), the sulphation reaction rate rose with increasing the temperature. On the contrary, in conditions of indirect sulphation (calcining conditions), the sulphation reaction rate increased with increasing the temperature up to 900-925 °C and then decreased probably due to the sinterization of the sorbent. As a consequence, the maximum sorbent conversion was reached at temperatures about 900-925 °C.

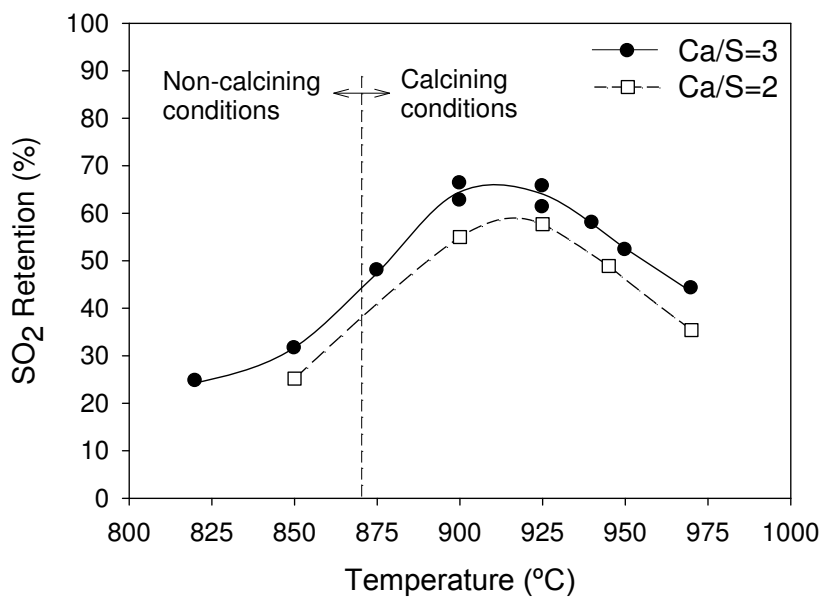


Fig. 3. Effect of temperature and Ca/S molar ratio on the sulphur retention with Granicarb limestone. O_2/CO_2 : 35/65. Calcining and non-calcining conditions correspond to thermodynamic data calculated with the gas inlet composition.

The SR values showed in Fig. 3 were obtained with residence times of ≈ 2 h. However, average residence times in a CFBC are clearly much higher than those used here, with values as high as 10-20 h (Lyngfelt and Leckner, 1992). It has to be also considered that the typical particle sizes used in CFBCs are smaller than those used in our installation (Anthony and Granaststein, 2001). Therefore, the SR values obtained in a CFBC could be very much higher than these showed in Fig. 3. This would be in concordance with some previous works on sorbent sulphation with long reaction times (Abanades et al., 2000), who demonstrated the high importance of the residual activity of the sorbents after pore plugging and the effect of the particle size.

Influence of the Ca/S molar ratio

As it is well known, the utilisation of Ca-based sorbents for SO_2 retention in FB boilers is not complete due to the pore blockage produced by CaSO_4 formation. In the typical operating conditions used in FB combustors, the sulphation reaction usually takes place at the external surface and around the pores of the sorbent particles. Since the molar volume of CaSO_4 is higher than the molar volume of CaCO_3 or CaO , the pores are blocked and the inner part of the particles remains essentially unsulphated (Anthony and Granaststein, 2001). Therefore, an important parameter in the SO_2 retention process is the Ca/S molar ratio used in the plant. This Ca/S molar ratio has to be necessarily higher than 1 in order to achieve a significant retention.

Fig. 3 compares the sulphur retention achieved in the oxy-fuel plant working with the anthracite at different Ca/S molar ratios. As expected, an increase in the Ca/S molar ratio produced higher sulphur retentions in all range of temperatures studied. It is also observed that the effect of the temperature was similar for the two different Ca/S molar ratios used, with maximum values at 900-925 °C.

Oxy fuel combustion vs. air combustion

One particular benefit of oxy-fuel firing is the ability to reuse technology suitable for existing pulverised coal fired power stations (Buhre et al., 2005; Wall et al., 2009). Typical differences for oxy-fired conditions compared to air-fired conditions are the higher O_2 content in the burner feed gas; the lower thermal efficiency, due to ASU and CO_2 compression energy requirements; the higher proportions of CO_2 and H_2O in the flue gas; the longer residence time of flue gas and fly ash in the furnace due to a lower volumetric gas flow ($\approx 80\%$ air fired); the higher dust concentration; the possibly different fly ash particle size; the lower NO_x production; and the increased SO_3/SO_2 concentration. These differences are mainly associated with the differences in gas properties and the effect of recycling the flue gas (Stanger and Wall, 2011).

In order to compare the effect on the SO_2 retention during oxy-fuel or air combustion conditions, new tests were carried out. For a better comparison, enriched-air conditions were selected to use the same O_2 concentration in both cases. Fig. 4 shows the results of the SO_2 retention obtained with the Granicarb limestone working with the anthracite at these conditions. The maximum SO_2 retention obtained under enriched-air combustion was obtained at 850-870 °C, while under oxy-fuel combustion the maximum retention was obtained at 925 °C. In addition, considering the optimal temperature for each case, the SO_2 retention obtained in oxy-fuel combustion is lower than that obtained in enriched-air combustion conditions.

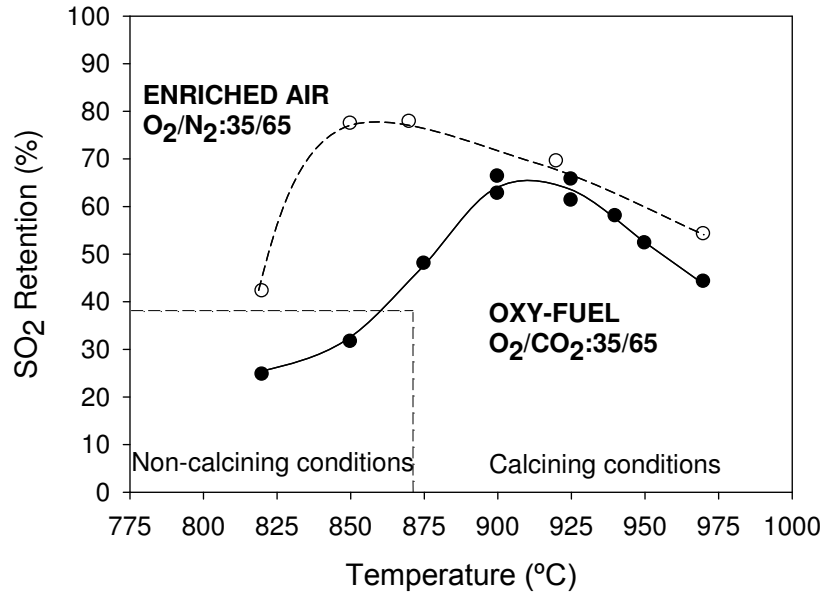


Fig. 4. Effect of combustion atmosphere on the sulphur retention with Granicarb limestone. Ca/S=3. Calcining and non-calcining conditions calculated with the gas inlet composition and thermodynamic data.

These results are in concordance with other authors who affirm the existence of a maximum sulphur capture efficiency at about 850 °C working in atmospheric FBC systems under air combustion conditions (O'Neil et al., 1979; Leckner et al., 1987). Anthony and Granatstein (2001) have summarised the explanations given by different authors to explain that maximum in the sulphur capture efficiency. According to the results obtained in this work, it seems that the maximum moves from ≈850 in conventional air-fired FBC systems to 900-925 °C under oxy-fuel combustion conditions.

CONCLUSIONS

- In oxy-fuel combustion in fluidised beds with limestone addition, the retention of SO₂ was higher working at calcining (indirect sulphation) conditions than at non-calcining (direct sulphation) conditions.
- The optimum FB combustor temperature from of point of view of sulphur retention shifted from ≈850 °C in conventional combustion with air to 900-925 °C in oxy-fuel combustion conditions.
- For Granicarb limestone, working at the optimum temperature for each technology, the SO₂ retentions obtained in oxy-fuel combustion conditions were slightly lower than those obtained under enriched-air combustion conditions.

NOMENCLATURE

$F_{0,coal}$	Feed rate of coal, kg/h
$X_{S,coal}$	Weight fraction of sulphur on coal, -
M_{SO_2}	Molar weight of SO ₂ , kg/kmol
Q	Gas flow rate, m ³ /h
C_{SO_2}	SO ₂ concentration, kmol/m ³
P_{CO_2}	CO ₂ partial pressure, kPa

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